

Dimensions

Steinitz Theorem

V vector space over K .

$X = (x_1, \dots, x_m)$ linearly independent list of vectors of V .

$Y = (y_1, \dots, y_m)$ system of generators of V .

a) $m \leq n$

b) m vectors of Y can be replaced by the vectors of X .
 $\Rightarrow$ system of generators for V .

! Not necessarily the first m vectors $\leadsto$ any m vectors.

Theorem

Any two bases of a vector space have the same nr. of elements.

Dimension of $V = \dim_K V = \dim V =$ the nr. of elements of any of its bases.

If $V = \{0\}$, then V has the basis $\emptyset$ and $\dim V = 0$.

Ex: a) K field

Canonical vector space: $\dim_K K^m = m$.

(*) 2 bases $\rightarrow$ same nr. of elem.

find a basis $\rightarrow$ nr. of elem. = dim.

$(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, \dots, 1)$

b) $\mathbb{R}^3 \rightarrow \dim = 3$.

subspaces

$\hookrightarrow$ any line passing through the origin $\rightarrow \dim = 1$.

$\hookrightarrow$ any plane passing through the origin $\rightarrow \dim = 2$.

$\hookrightarrow \mathbb{R}^3$ itself $\rightarrow \dim = 3$.

$\vec{i}, \vec{j}, \vec{k}$ unit vectors on each axis.

c) $K_m[X]$ the set of polynomials of degree at most m :

$(1, x, x^2, \dots, x^m)$ basis $\Rightarrow m+1$ elem. $\Rightarrow \dim = m+1$.

d) $M_2(K)$

$$\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right) \rightarrow \dim = 4.$$

e) $S = \{ (x, y, z) \in \mathbb{R}^3 \mid x - y - z = 0 \} = \langle (1, 1, 0), (1, 0, 1) \rangle$
 $x = y + z$

$$\alpha_1 (1, 1, 0) + \alpha_2 (1, 0, 1) = 0$$

$$\Leftrightarrow \begin{cases} \alpha_1 + \alpha_2 = 0 \\ \alpha_1 = 0 \\ \alpha_2 = 0 \end{cases} \Rightarrow \text{independent.}$$

f) $\mathbb{C} \rightarrow \{1\}$ basis $\rightarrow \dim = 1$

$\mathbb{R}\mathbb{C} \rightarrow \{1, i\}$ basis $\rightarrow \dim = 2$

Theorem

V vector space over K .

a) $\dim V = m$

b) the max. no of linearly indep. vectors in V is m .

c) the min. no of generators for V is m .

Theorem (*)

V vector space over K

$\dim V = m$

$X = (u_1, \dots, u_m)$ list of vectors in V .

$\Rightarrow X$ linearly indep.
 $(\Leftarrow)$

X system of generators.

" $\Leftarrow$ " ... $V = \langle X \rangle = \{u_1, \dots, u_{j-1}, u_{j+1}, \dots, u_n\}$ (both conditions for basis)

$\hookrightarrow m-1, \text{ not } m \Rightarrow \text{ctr. } (u_j \rightarrow u_i)$

$\dim_{\mathbb{R}} \mathbb{R}^3 = 3$

Show that $(1, 1, 1), (1, 1, 0), (1, 0, 0)$ is another basis for $\mathbb{R}^3$.

$H_1 \rightarrow$ def. basis

$H_2 \rightarrow$ write as a linear comb.

H_3

theorem (*) $\rightarrow$ you may show that it is lin. indep.

Justify like this
for the exam!

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} \neq 0 \Rightarrow \text{indep.}$$

We have 3 linearly indep. vectors in a vector space of dim 3.
 $\Rightarrow$ theorem (*) $\Rightarrow$ they are also a system of generators.
 $\Rightarrow$ basis.

Corollary:

n vectors in K^n form a basis of the canonical vector space K^n

$(\Leftrightarrow)$ they are linearly indep.

$(\Leftrightarrow)$ the det. consisting of their components is $\neq 0$.

Theorem:

Any linearly independent list of vectors in a vector space can be completed to a basis of the vector space.

Corollary:

V vector space over K and $S \subseteq V$

a) Any basis of S is a part of a basis of V

b) $\dim S \leq \dim V$

c) $\dim S = \dim V \Leftrightarrow S = V$

use
(*) for proof

basis not unique $\Rightarrow$ completion not unique

the list (e_1, e_2) , where $e_1 = (1, 0, 0)$ is linearly indep. in $\mathbb{R}^3$
 $e_2 = (0, 1, 0)$ (the canonical vector space)

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

$\swarrow$ complete with $e_3 = (0, 0, 1) \Rightarrow$ canonical basis.

Decomposition theorem

It can be

Theorem:

V vector space over K .

$$S \leq V$$

$$\Rightarrow \exists \bar{S} \leq V \text{ s.t. } V = S \oplus \bar{S}.$$

In particular: $\dim V = \dim S + \dim \bar{S}$.

$$V = S \oplus \bar{S} : \begin{cases} V = S + \bar{S} \\ S \cap \bar{S} = \{0\} \end{cases} \quad \text{or } (\forall) v \in V, \exists! s \in S \text{ s.t. } v = s + \bar{s} \text{ } s \in S$$

DIRECT SUM



it allows us to break down stuff into "smaller" subspaces

Complement of a subspace

$$\left. \begin{array}{l} V \text{ vector space over } K \\ S \leq V \end{array} \right\} \Rightarrow \begin{array}{l} \bar{S} \text{ subspace of } V \text{ s.t. } V = S \oplus \bar{S} \\ \text{"COMPLEMENT of } S \text{ in } V. \end{array}$$

RECIPE

take subspace
complete
extra vectors
take them
subspace generated by them

Theorem

V, V' vector spaces over K

$$V \cong V' \Leftrightarrow \dim V = \dim V'$$

! Any isomorphism preserves BASES.

- any inj. linear map preserves linear independence
- any surj. linear map preserves spanning sets
- any bij. linear map preserves $\left\{ \begin{array}{l} \text{linear independence} \\ \text{spanning sets} \end{array} \right\}$ bases.

! Theorem:

Any vector space V over K with $\dim V = m$ is $\cong$ to the canonical one.



"STRUCTURE RESULT"

First Dimension Theorem

$$\left\{ \begin{array}{l} f(v_1 + v_2) = f(v_1) + f(v_2) \\ f(k \cdot v_1) = k \cdot f(v_1) \end{array} \right.$$

Def: $f: V \rightarrow V'$ is a K -linear map.

- 1) $\dim(\text{Ker } f) = \text{"nullity of } f" = \text{null}(f)$
- 2) $\dim(\text{Im } f) = \text{"rank of } f" = \text{rank}(f)$

Theorem: $f: V \rightarrow V'$ is a K -linear map.

Then

$$\dim V = \dim(\text{Ker } f) + \dim(\text{Im } f)$$

$$\dim V = \text{null}(f) + \text{rank}(f)$$

Second Dimension Theorem

V vector space over K

S, T subspaces of V .

$$\dim S + \dim T = \dim(S \cap T) + \dim(S + T)$$

Corollary

V vector space over K

S, T subspaces of V

$$V = S \oplus T$$

$$\dim V = \dim S + \dim T$$

In the exam you may be asked:

2 subspaces S, T . Their dimensions?

$$\dim(S + T) = \dim(\langle S \cup T \rangle)$$

$$\hookrightarrow S + T = \langle S \cup T \rangle$$

Def: DOT-PRODUCT / SCALAR PRODUCT.

$$u = (x_1, \dots, x_m) \in K^m$$

$$v = (y_1, \dots, y_m) \in K^m$$

$$u \cdot v = x_1 \cdot y_1 + x_2 \cdot y_2 + \dots + x_m \cdot y_m \in K \rightarrow \text{dot product.}$$

